

Roll No.

TPE-505

B. TECH. (PE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
PETROLEUM REFINING ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What are the refineries feedstocks ? Classify the different physical and chemical processes involved in the refining of crude oil. 10 Marks (CO1)

OR

- (b) What do you mean by characterization factor developed by UOP ?
Discuss the classification of crude oils. 10 Marks (CO1)

2. (a) On which laboratory tests basis the crude oil characterization is done ?
Explain on the basis of volatility combustion and viscosity in detail.

10 Marks (CO2)

OR

- (b) Explain petroleum refining, operation and optimization. Discuss the selection of processes for optimization. 10 Marks (CO2)

P. T. O.

(2)

3. (a) Define the following :

10 Marks (CO2)

- (i) Octane number
- (ii) Cetane number
- (iii) Performance number
- (iv) Diesel index
- (v) Viscosity index

OR

(b) On which basis the classification of continuous distillations columns is categorized ? Discuss.

10 Marks (CO2)

4. (a) Draw the flow sheet diagram of crude oil desalting process and discuss its process description stepwise.

10 Marks (CO1, CO2)

OR

(b) Discuss the main components of distillation columns with the help of a neat and labeled diagram.

10 Marks (CO1, CO2)

5. (a) Draw the flow sheet diagram of vacuum distillation of reduced crude oil (VDU). Explain it with the process description.

10 Marks (CO2)

OR

(b) Explain the function of the following in the distillation column :

10 Marks (CO2)

- (i) Reflux
- (ii) Reboiler/Stripping steam
- (iii) Pulsation
- (iv) Dumping